

Houlin Wang

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HIGHLIGHTS

Zhejiang University BME with CS minor | GPA: 91.25/100 | IELTS: 8.0 | Biomedical AI · iEEG · Embedded Systems

RESEARCH PROFILE

Biomedical Engineering undergraduate at Zhejiang University with research experience in intracranial EEG analysis, medical image AI, biomedical sensing, and embedded instrumentation. Interested in developing reproducible machine-learning and signal-processing methods for biomedical data, bridging computational analysis with sensing hardware and embedded medical systems.

RESEARCH AREAS

Biomedical AI and medical image analysis; biomedical signal processing and neural engineering; biosensor instrumentation and embedded medical systems.

EDUCATION

Zhejiang University

B.Eng. in Biomedical Engineering (Instrumentation); Minor in Computer Science

Sep. 2023 – Expected Jun. 2027

GPA: 91.25/100 (4.12/4.3)

- Interdisciplinary training across biomedical engineering, computer science, embedded systems, signal processing, and medical foundations.
- Completed 188 credits in the first three undergraduate years, representing the highest accumulated course load in the major.
- Selected coursework: Machine Learning, Biomedical Signal Processing, Biomedical Image Processing, Biomedical Sensors and Measurement, Instrument System Design, Embedded Systems, Circuits and Electronics, Data Structures, Digital Logic Design, Hardware Description Language, Probability and Statistics.
- Medical preparation: audited Regional Anatomy and participated in anatomy laboratory sessions; enrolled in an English-taught Systematic Anatomy course.

RESEARCH EXPERIENCE

Medial Temporal Lobe iEEG Analysis of Working Memory and Hippocampal Ripples

Undergraduate Researcher, Qiushi Academy for Advanced Studies, Zhejiang University

Aug. 2025 – Present

Supervisor: Prof. Shaomin Zhang

- Designed and implemented a reproducible MATLAB pipeline for hippocampal-ripple detection and trial-level working-memory analysis across 9 iEEG participants, 37 clean LFP sessions, 90 valid contacts, and 1,683 clean trials.
- Tested how ripple dynamics varied across task phases, memory loads, and behavioral outcomes, and analyzed co-ripple patterns and event-triggered activity using permutation-based controls and exploratory statistical analyses.
- Established cross-session trial mapping, event-alignment, hemisphere-matching, and quality-control procedures to ensure data integrity and valid cross-region event matching.
- Developed trial-specific reinstatement and matched non-ripple controls; exploratory results suggest stronger same-trial anterior-posterior hippocampal coordination during successful memory maintenance.

AI-Based Corneal Confocal Microscopy Dataset and Diagnosis System

Undergraduate Researcher, Zhejiang University

Apr. 2025 – May 2026

Supervisor: Prof. Yingke Xu

- Contributed to an in vivo corneal confocal microscopy dataset and diagnosis project comprising 1,524 images, including 1,042 pathological and 482 normal images across 11 categories.
- Trained and debugged classification models, performing image preprocessing, model evaluation, and class-wise performance analysis across normal and pathological categories.
- Contributed to CLIP-based feature experiments, multi-scale feature fusion, and semantic-segmentation evaluation within the broader medical-image learning workflow.
- Project models achieved 96.85% mean classification accuracy and 98.2% mean AUC across 11 image categories, while the segmentation model achieved a mean Dice score of 82.27%.
- Co-authored the associated dataset manuscript, currently under review at *Scientific Data*.

PUBLICATIONS

- Jiang, C., Yang, R., **Wang, H.**, et al. "A Multi-Disease In Vivo Confocal Microscopy Dataset for AI-Based Corneal Disease Diagnosis and Research." Submitted to *Scientific Data*, under review, 2026.

ENGINEERING AND COMPETITION EXPERIENCE

- Vision-Guided Lissajous Figure Display and Control System

Jun. 2026 – Aug. 2026

Competition Team Member, Zhejiang University; National Undergraduate Electronics Design Contest & TI Cup

 - Developed a hardware-software prototype for tracking unknown 1–100 kHz sinusoidal inputs and generating phase-controlled same-frequency or double-frequency outputs for oscilloscope XY-mode Lissajous display.
 - Designed and debugged signal-conditioning and output-control modules, including attenuation, DC biasing, analog filtering, variable-gain amplification, STM32H750-based signal detection, and AD9910 DDS control.
 - Built the vision-recognition workflow by generating 50,000 simulated oscilloscope images, training and deploying a MobileNetV3 classifier, and adding multi-round voting for more stable pattern identification.
 - Integrated K230 and Raspberry Pi 5 visual feedback with binary-search frequency refinement, local-scan fallback, and phase-adjustment logic to support time-constrained automatic tuning of Lissajous patterns.

- ZJU TruSense Biosensor Design Competition Team

Oct. 2024 – Sep. 2025

Instrumentation and Hardware Team Member, Zhejiang University; SensUs Student Competition

 - Developed the instrumentation subsystem of a portable creatinine-sensing platform for acute kidney injury monitoring.
 - Participated in hardware-circuit design and implemented a three-electrode electrochemical measurement circuit for acquiring creatinine-dependent sensor signals.
 - Led PCB layout, fabrication, and five hardware iterations for an MSP430FR2433-based system, reducing the prototype to an approximately 2.5 cm-diameter form factor.
 - Built and demonstrated the physical hardware prototype through soldering, board-level debugging, system integration, and partial embedded code development.
 - Project recognized with Final Medal, Public Inspiration 1st Prize, and Innovation 3rd Prize at SensUs Student Competition 2025.

TECHNICAL SKILLS

Programming	C++/C, MATLAB, Python, NumPy, pandas, PyTorch, Java, Git
Biomedical Data	FieldTrip, EEGLAB, FMA Toolbox; permutation-based controls, exploratory statistical analyses
Medical AI	Classification, semantic segmentation, CLIP-based encoding, multi-scale feature fusion, error analysis
Embedded Hardware	STM32, MSP430/MSPM0, ESP32, 8051; electrochemical sensing, analog/digital circuit design, signal conditioning, PCB schematic and layout
Digital Toolchains	Verilog, Vivado, CCS, Keil, STM32CubeMX, EasyEDA
Documentation	LaTeX, Typst, Markdown

HONORS AND AWARDS

- Scholarships

 - Zhejiang Provincial Government Scholarship, 2024–2025.
 - Second-Class Scholarship, Zhejiang University, 2024–2025.
 - Third-Class Scholarship, Zhejiang University, 2023–2024.
- Competition Awards

 - Gold Medal, SynBio Challenges 2025, Red Track: Health, Medical & Treatment, Team TruSense, Aug. 2025.
 - National Bronze Award (Final Round) and Provincial First Prize (Provincial Round), 11th National College Student Medical Innovation Competition and “Belt and Road” International Competition, Interdisciplinary Track, Jun.–Aug. 2025.
 - Final Medal; Public Inspiration 1st Prize; Innovation 3rd Prize, SensUs Student Competition 2025: Kidney Failure, Team TruSense, Aug. 2025.
 - Third Prize, 10th National College Student Biomedical Engineering Innovation Design Competition, Jul. 2025.

LANGUAGES

Chinese	Native
English	IELTS 8.0 (R9.0 / L9.0 / W7.0 / S6.5); TOEFL iBT 101 (R29 / L24 / S25 / W23)
French	Simple conversational ability; TFU-4 96/100

VOLUNTEER EXPERIENCE

- Completed 260+ hours of volunteer service.
- Volunteer Teacher, Western China Education Volunteer Program, Hongya, Sichuan, Jul.–Aug. 2024; served as a homeroom teacher and instructor.